# Aravind Rajeswaran

Curriculum Vitae - August 2026
Website: https://aravindr93.github.io

## Employment

**Microsoft AI** - Principal Researcher *July 2025 - present*
Grounding, SFT, and RL for computer use agents (CUA)
Notable Projects: Fara1.5, Fara-7B

**Meta Platforms (FAIR)** - Research Scientist *April 2021 - April 2025*
Foundation models and benchmarks for Embodied AI
Notable Projects: OpenEQA, VC-1, R3M, RoboHive

**Google Brain** - Research Internship *June 2019 - May 2020*
Model-based reinforcement learning for robotics
Notable Projects: MOReL, MBRL-Game

**OpenAI** - Research Internship *June 2017 - Sep 2017*
Algorithms for dexterous robot hands and multi-agent RL
Notable Projects: DAPG

## Education

**University of Washington Seattle** *Sep 2016 – June 2021*
Ph.D. in Computer Science & Engineering
Advisers: Profs. Sham Kakade and Emo Todorov

**Indian Institute of Technology Madras**, BTech (Hons.) *Aug 2011 – July 2015*
Advisers: Profs. Balaraman Ravindran and Shankar Narasimhan

## Academic Awards

- Best paper award at the ICRA 2022 Scaling Robot Learning Workshop *2022*
- Best paper award finalist at the RSS 2022 Scaling Robot Learning Workshop *2022*
- J. P. Morgan PhD Fellowship in AI *2020*
- Facebook PhD fellowship finalist in ML *2020*
- Best paper award at IEEE SIMPAR *2018*
- University of Washington PhD fellowship *2016*
- Bhagyalakshmi and Krishna Ayengar award for best undergraduate thesis. *2015*

## Publications

### Preprints

[1] *Fara-1.5: Scalable Learning Environments for Computer Use Agents*
*Note: alphabetical author ordering; Role: overall model training lead*
A. Awadallah, S. Gupta, Y. Lara, Y. Lu, H. Mozannar, A. Nambi, Z. Nussbaum, Y. Pandya, A. Rajeswaran, C. Rosset, A. Taymanov, L. do Valle, V. Vineet, S. Whitehead, A. Zhao
arXiv Preprint, 2026.

### Conference Articles

(* denotes equal first-author contribution, † denotes equal senior-authorship for leadership/advising)

[2] *LOCATE 3D: Real-World Object Localization via Self-Supervised Learning in 3D*
P. McVay, S. Arnaud, A. Martin, A. Majumdar, K.M. Jatavallabhula, P. Thomas, R. Partsey, D. Dugas, A. Gejji, A. Sax, V.-P. Berges, M. Henaff, A. Jain, A. Cao, I. Prasad, M. Kalakrishnan, M. Rabbat, N. Ballas, M. Assran, O. Maksymets, A. Rajeswaran†, F. Meier†
International Conference on Machine Learning (**ICML**), 2025. **(Spotlight)**

[3] *OpenEQA: Embodied Question Answering in the Era of Foundation Models*
A. Majumdar, A. Ajay, X. Zhang, P. Putta, S. Yenamandra, M. Henaff, S. Silwal, P. McVay, O. Maksymets, S. Arnaud, K. Yadav, Q. Li, B. Newman, M. Sharma, V. Berges, S. Zhang, P. Agrawal, Y. Bisk, D. Batra, M. Kalakrishnan, F. Meier, C. Paxton, A. Sax, A. Rajeswaran
Computer Vision and Pattern Recognition (**CVPR**), 2024.

[4] *MoDem-V2: Visuo-Motor World Models for Real-World Robot Manipulation*
P. Lancaster, N. Hansen, A. Rajeswaran, V. Kumar
International Conference on Robotics and Automation (**ICRA**), 2024.

[5] *From LLMs to Actions: Latent Codes as Bridges in Hierarchical Robot Control*
Y. Shentu, P. Wu, A. Rajeswaran, P. Abbeel
International Conference on Intelligent Robots and Systems (**IROS**), 2024.

[6] *RoboHive: A Unified Framework for Robot Learning*
V. Kumar, R. Shah, G. Zhou, V. Moens, V. Caggiano, A. Gupta, A. Rajeswaran
Advances in Neural Information Processing Systems (**NeurIPS**), 2023.

[7] *Where are we in the search for an Artificial Visual Cortex for Embodied Intelligence?*
A. Majumdar, K. Yadav, S. Arnaud, J. Ma, C. Chen, S. Silwal, A. Jain, V.-P. Berges, T. Wu, J. Vakil, P. Abbeel, J. Malik, D. Batra, Y. Lin†, O. Maksymets†, A. Rajeswaran†, F. Meier†
Advances in Neural Information Processing Systems (**NeurIPS**), 2023.

[8] *Masked Trajectory Models for Prediction, Representation, and Control*
P. Wu, A. Majumdar, K. Stone, Y. Lin, I. Mordatch, P. Abbeel, A. Rajeswaran
International Conference on Machine Learning (**ICML**), 2023.

[9] *On Pre-Training for Visuo-Motor Control: Revisiting a Learning-from-Scratch Baseline*
N. Hansen, Z. Yuan, Y. Ze, T. Mu, A. Rajeswaran†, H. Su†, H. Xu†, X. Wang†
International Conference on Machine Learning (**ICML**), 2023.

[10] *MoDem: Accelerating Visual Model-Based Reinforcement Learning with Demonstrations*
N. Hansen, Y. Lin, H. Su, X. Wang, V. Kumar, A. Rajeswaran
International Conference on Learning Representations (**ICLR**), 2023.

[11] *Real World Offline Reinforcement Learning with Realistic Data Source*
G. Zhou, L. Ke, S. Srinivasa, A. Gupta, A. Rajeswaran, V. Kumar
International Conference on Robotics and Automation (**ICRA**), 2023.

[12] *R3M: A Universal Visual Representation for Robot Manipulation*
S. Nair, A. Rajeswaran, V. Kumar, C. Finn, A. Gupta
Conference on Robot Learning (**CoRL**), 2022.

[13] *The (Un)Surprising Effectiveness of Pre-Trained Vision Models for Control*
S. Parisi*, A. Rajeswaran*, S. Purushwalkam, A. Gupta
International Conference on Machine Learning (**ICML**), 2022. **(Long Oral)**

[14] *CIC: Contrastive Intrinsic Control for Unsupervised Skill Discovery*
M. Laskin, H. Liu, X.B. Peng, D. Yarats, A. Rajeswaran, P. Abbeel
Advances in Neural Information Processing Systems (**NeurIPS**), 2022.

[15] *Can Foundation Models Perform Zero-Shot Task Specification For Robot Manipulation?*
Y. Cui, S. Niekum, A. Gupta, V. Kumar, A. Rajeswaran
Learning for Dynamics and Control (**L4DC**), 2022.

[16] *Decision Transformer: Reinforcement Learning via Sequence Modeling*
L. Chen, K. Lu, A. Rajeswaran, K. Lee, A. Grover, M. Laskin, P. Abbeel, A. Srinivas, I. Mordatch
Advances in Neural Information Processing Systems (**NeurIPS**), 2021.

[17] *Visual Adversarial Imitation Learning using Variational Models*
R. Rafailov, T. Yu, A. Rajeswaran, C. Finn
Advances in Neural Information Processing Systems (**NeurIPS**), 2021.

[18] *COMBO: Conservative Offline Model-Based Policy Optimization*
T. Yu, A. Kumar, R. Rafailov, A. Rajeswaran, S. Levine, C. Finn
Advances in Neural Information Processing Systems (**NeurIPS**), 2021.

[19] *Reinforcement Learning with Latent Flow*
W. Shang, X. Wang, A. Srinivas, A. Rajeswaran, Y. Gao, P. Abbeel, M. Laskin
Advances in Neural Information Processing Systems (**NeurIPS**), 2021.

[20] *Offline Reinforcement Learning from Images with Latent Space Models*
R. Rafailov, T. Yu, A. Rajeswaran, C. Finn
Learning for Dynamics and Control (**L4DC**), 2021.

[21] *MOReL: Model-Based Offline Reinforcement Learning*
R. Kidambi*, A. Rajeswaran*, P. Netrapalli, T. Joachims
Advances in Neural Information Processing Systems (**NeurIPS**), 2020.

[22] *A Game Theoretic Framework for Model Based Reinforcement Learning*
A. Rajeswaran, I. Mordatch, V. Kumar
International Conference on Machine Learning (**ICML**), 2020.

[23] *Lyceum: An Efficient and Scalable Ecosystem for Robot Learning*
C. Summers, K. Lowrey, A. Rajeswaran, S. Srinivasa, E. Todorov
Learning for Dynamics and Control (**L4DC**), 2020.

[24] *Meta-Learning with Implicit Gradients*
A. Rajeswaran*, C. Finn*, S. Kakade, S. Levine
Advances in Neural Information Processing Systems (**NeurIPS**), 2019.

[25] *Online Meta-Learning*
C. Finn*, A. Rajeswaran*, S. Kakade, S. Levine
International Conference on Machine Learning (**ICML**), 2019.

[26] *Plan Online, Learn Offline: Efficient Learning and Exploration via Model-Based Control*
K. Lowrey*, A. Rajeswaran*, S. Kakade, E. Todorov, I. Mordatch
International Conference on Learning Representations (**ICLR**), 2019.

[27] *Dexterous Manipulation with Deep Reinforcement Learning: Efficient, General, and Low-Cost*
H. Zhu, A. Gupta, A. Rajeswaran, S. Levine, V. Kumar
International Conference on Robotics and Automation (**ICRA**), 2019.

[28] *Learning Deep Visuomotor Policies for Dexterous Hand Manipulation*
D. Jain, A. Li, S. Singhal, A. Rajeswaran, V. Kumar, E. Todorov
International Conference on Robotics and Automation (**ICRA**), 2019.

[29] *Reinforcement Learning for Non-Prehensile Manipulation: Transfer from Simulation to Physical System.* K. Lowrey, S. Kolev, J. Dao, A. Rajeswaran, E. Todorov
IEEE International Conference on Simulation, Modeling, and Programming for Autonomous Robots (**SIMPAR**), 2018. **(Best Paper Award)**

[30] *Variance Reduction for Policy Gradient with Action-Dependent Factorized Baselines*
C. Wu, A. Rajeswaran, Y. Duan, V. Kumar, A. Bayen, S. Kakade, I. Mordatch, P. Abbeel
International Conference on Learning Representations (**ICLR**), 2018. **(Full Oral)**

[31] *Divide-and-Conquer Reinforcement Learning*
D. Ghosh, A. Singh, A. Rajeswaran, V. Kumar, S. Levine
International Conference on Learning Representations (**ICLR**), 2018.

[32] *Learning Complex Dexterous Manipulation with Deep Reinforcement Learning and Demonstrations*
A. Rajeswaran*, V. Kumar*, A. Gupta, G. Vezzani, J. Schulman, E. Todorov, S. Levine
Proceedings of Robotics: Science and Systems (**RSS**), 2018.

[33] *Towards Generalization and Simplicity in Continuous Control*
A. Rajeswaran*, K. Lowrey*, E. Todorov, S. Kakade
Advances in Neural Information Processing Systems (**NeurIPS**), 2017.

[34] *EPOpt: Learning Robust Neural Network Policies Using Model Ensembles*
A. Rajeswaran, S. Ghotra, B. Ravindran, S. Levine
International Conference on Learning Representations (**ICLR**), 2017.

### Journal Articles

[35] *Identifying Topology of Power Distribution Networks Based on Smart Meter Data*
S. Jayadev, N. Bhatt, R. Pasumarthy, A. Rajeswaran
IEEE Transactions on Smart Grid, 2017.

[36] *A Graph Partitioning Approach for Leak Detection in Water Distribution Networks*
A. Rajeswaran, S. Narasimhan, S. Narasimhan
Computers & Chemical Engineering, 2017.

## Media Coverage (Selected Subset)

- **VentureBeat** – *"Microsoft's Fara-7B is a computer-use AI agent that rivals GPT-4o and works directly on your PC"* ***2025***
- **VentureBeat** – *"Meta AI releases OpenEQA to spur 'embodied intelligence'..."* ***2024***
- **Yahoo Finance** – *"Meta is teaching robots how to move on their own"* ***2023***
- **SiliconANGLE** – *"Meta's AI researchers create artificial visual cortex for robots..."* ***2023***
- **ZDNet** – *"A Berkeley mash-up of AI approaches promises continuous learning"* ***2019***

## Mentoring

### PhD Interns

- Philipp Wu (UC Berkeley, PhD → startup)
- Arjun Majumdar (GATech, PhD → Meta)
- Anurag Ajay (MIT, PhD → DeepMind)
- Mandi Zhao (Stanford, PhD)
- Nicklas Hansen (UCSD, PhD)
- Allan Zhou (Stanford, PhD → OpenAI)
- Suraj Nair (Stanford, PhD → PI)
- Liyiming Ke (UW, PhD → PI)
- Andi Peng (MIT, PhD → startup)
- Yuchen Cui (UT Austin, PhD → UCLA)

### University Students & AI Residents

- Shikhar Bahl (CMU, PhD → Skild.AI)
- Aryan Jain (UC Berkeley, BS → Tesla)
- Sergio Arnaud (AI Resident, FAIR → Meta)
- Rafael Rafailov (Stanford, MS → Reflection AI)
- Karmesh Yadav (FAIR → Georgia Tech, PhD)
- Ben Evans (UW, BS/MS → NYU, PhD)
- Sarvjeet Ghotra (IITM Intern → MILA, PhD)
- Colin Summers (UW, BS/MS → UW, PhD)
- Catherine Cang (Berkeley, BS → Reflection AI)
- Divye P. Jain (UW, BS/MS → Google)

## Professional Service and Teaching

### Course Instructor and TA

- Fully designed and **taught** a special topics course at UW on deep RL for robotics. [course website]
- Teaching assistant for advanced graduate level machine learning courses at UW.

### Workshops Organized

- Pretraining for Robot Learning (website), CoRL 2022.
- 3rd Offline RL workshop: Offline RL as a "Launchpad" (website), NeurIPS 2022.
- Object Representations for Learning and Reasoning (website), NeurIPS 2020.
- Generative Modeling and Model-Based Reasoning for Robotics and AI (website), ICML 2019.

### Peer Review Service (Selected Subset)

- IEEE/CVF Conference on Computer Vision and Pattern Recognition – CVPR (2023)
- Neural Information Processing Systems – NeurIPS (2018, 2021, 2023, 2025, 2026)
- International Conference on Robotics and Automation – ICRA (2019, 2021)
- IEEE Robotics and Automation Letters (RA-L)